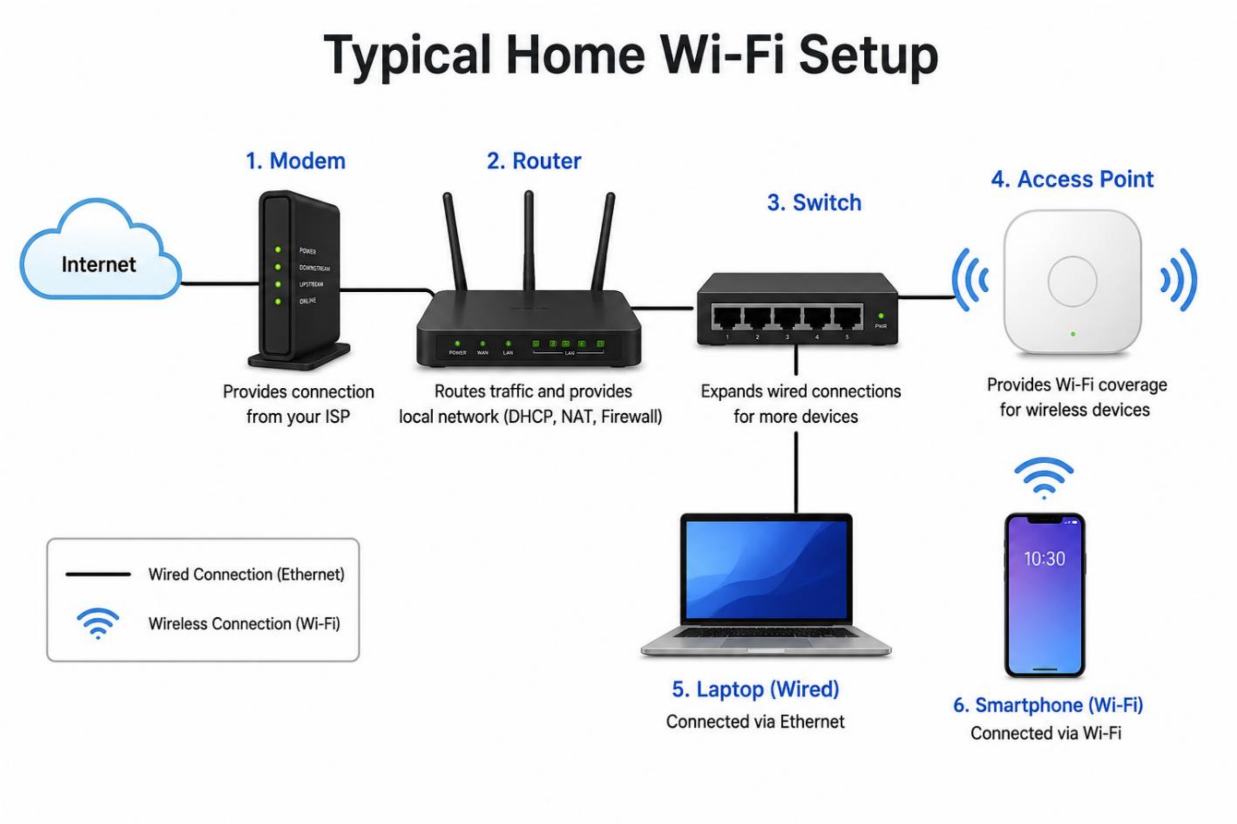
1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2.List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Hardware Device	Main Function	Real-Life Example
Modem	Connects your home or office network to the Internet Service Provider (ISP) and converts the internet signal into a usable form.	At home, a modem connects the internet line from the ISP (such as JioFiber or Airtel Xstream) to your network.
Router	Distributes the internet connection to multiple devices and manages network traffic. It also provides Wi-Fi connectivity.	At home, a router allows laptops, smartphones, smart TVs, and other devices to share the same internet connection.
Switch	Connects multiple wired devices within the same Local Area Network	In an office or computer lab, a switch connects many desktop computers,

Networking Hardware Device	Main Function	Real-Life Example
	(LAN) and transfers data efficiently between them.	printers, and servers using Ethernet cables.
Access Point (AP)	Extends or creates a wireless (Wi-Fi) network, allowing wireless devices to connect to the LAN.	In schools, colleges, hotels, or large offices, access points provide Wi-Fi coverage across different rooms or floors.
Network Cable (Ethernet Cable)	Physically connects networking devices and provides a fast, stable wired network connection.	At home or in an office, an Ethernet cable connects a PC, gaming console, or switch directly to the router for a reliable internet connection.

3.Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Network Cable Type	Common Use	Source
Ethernet Cable (RJ45/Cat6)	Commonly used to connect computers, routers, switches, and gaming consoles for fast and reliable wired networking.	https://en.wikipedia.org/wiki/Ethernet_over_twisted_pair
Coaxial Cable	Commonly used for cable TV connections, cable internet, and connecting a modem to the ISP's network.	https://en.wikipedia.org/wiki/Coaxial_cable
Fiber Optic Cable	Commonly used by Internet Service Providers (ISPs) and businesses to deliver very high-speed internet over long distances.	https://en.wikipedia.org/wiki/Optical_fiber_cable

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	**Router**	**Switch**
Main Use	Connects different networks (such as a home network to the Internet).	Connects multiple devices within the same Local Area Network (LAN).
Typical Location	Found at homes, offices, and schools to provide internet access.	Found in offices, computer labs, and server rooms to connect multiple wired devices.
How They Handle Data	Uses **IP addresses** to route data between different networks.	Uses **MAC addresses** to forward data only to the intended device.
Internet Connection	Connects directly to the modem and shares internet with all devices.	Does not connect directly to the internet; it expands the number of wired network connections.
Network Coverage	Can provide both wired and wireless (Wi-Fi) connectivity.	Provides only wired Ethernet connections (unless connected to an access point).
OSI Layer	Operates mainly at **Layer 3 (Network Layer)**.	Operates mainly at **Layer 2 (Data Link Layer)**.